MIL-F-82191A(MC)
29 December 1978
SUPERSEDING
MIL-F-82191(MC)
19 August 1968

MILITARY SPECIFICATION

FUEL HOSE EVACUATION KITS

This specification is approved for use by U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fuel hose evacuation kits.

1.2 Classification. The fuel hose evacuation kits shall be of the following sizes, as specified in 6.2.1:

Sizes

Size 1 - For use with 4-inch fuel hose.
Size 2 - For use with 6-inch fuel hose.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4930

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SPECIFICATIONS

FEDERAL

MMM-A-122	- Adhesive, Butadiene Acrylonitrile Base, Medium Solids, General Purpose.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-S-7916	- Sealing Compound, Thread and Gasket, Fuel, Oil, and Water Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

U.S. MARINE CORPS

DL77C5208A0032	- Fuel Hose Evacuation Kit, 4-Inch.
DL77C5208A0046	- Fuel Hose Evacuation Kit, 6-Inch.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit(s) for first article inspection and approval (see 4.4 and 6.3).

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3.2 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3 Design and construction. The fuel hose evacuation kits shall be in accordance with Data List DL77C5208A0032 for size 1 and Data List DL77C5208A0046 for size 2, and as specified herein. These drawings are end product drawings and no deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the fuel hose evacuation kits. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor, or obtained from a vendor, to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.3.1 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.3.2 Castings and forgings. Cast and forged components of the fuel hose evacuation kit shall be sound and free from blowholes, hard spots, porosity, and cracks. Castings shall not be repaired, plugged, brazed, or burned. All fins and roughness shall be removed.

3.3.3 Metal fabrication. Metal used in the fabrication of the fuel hose evacuation kit shall be free from kinks, ripples, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Methods of cutting that utilize heat shall not be used for aluminum. Sheared, chipped and sawed edges shall be free from burrs and slivers. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating and heated metal shall be allowed to cool slowly. Boltholes shall be accurately formed and have burrs removed. External surfaces shall be free from sharp edges and corners except where sharp edges or corners are required or where they are not detrimental to safety.

3.3.4 Threaded connections and fasteners. All threaded connections and fasteners shall show no evidence of cross threading, multilation, or

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detrimental or hazardous burrs. The length of screws and bolts installed with nuts shall be such that the exposed portion is a minimum of two threads. Maximum lengths shall be limited by the nearest standard screw length. All threaded fasteners shall be tight. No relative movement shall be possible between the attached parts. Threaded fasteners shall be sealed with sealing compound conforming to MIL-S-7916.

3.3.5 Cleaning. The fuel hose evacuation kit shall be cleaned and dried of all cutting oil, grease, residue of tests, dirt, metal chips, and other extraneous material in accordance with MIL-P-116.

3.4 Performance characteristics. The fuel hose evacuation kit shall have the operating characteristics specified on the applicable drawings (reference drawings 77C5208A0052, 77C5208A0055, and 77C5208A0057) and as specified herein (see 4.6.2.1 and 4.6.2.2).

3.4.1 Rough handling. The fuel hose evacuation kit shall be capable of being dropped without sustaining cracks, stress damage, or permanent deformation (see 4.6.2.3).

3.5 Identification and instruction plates.

3.5.1 Identification marking. Each fuel hose evacuation kit shall be identified in accordance with MIL-STD-130. The marking shall be applied on plates conforming to MIL-P-514, type I, style 3, composition C, of type optional, grade B, class 1 material. The plates shall be attached to the outside of the chest by adhesive, screws, rivets, or bolts in a place where the plate will be visible and legible. Adhesive, if used, shall conform to MMM-A-122.

3.5.2 Instruction plate. Unless otherwise specified (see 6.2.1), each fuel hose evacuation kit shall be equipped with an instruction plate conforming to MIL-P-514, type III, composition C, of type optional, grade B, class 1 material. The plate shall include all warnings and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the fuel hose evacuation kit. The plate shall be attached to the chest by adhesive, screws, rivets, or bolts in a place where the plate will be visible and legible. Adhesive, if used, shall conform to MMM-A-122.

3.6 Workmanship. All parts, components and assemblies of the fuel hose evacuation kit shall be free from extraneous material such as sand, dirt, pits, sprues, scale, flux, grease, or oils that would render the fuel hose evacuation kit unsuitable for its intended use. All parts, components, and assemblies of the fuel hose evacuation kit shall be free from misalinement, incomplete welds, leaks, or other defects that could impair the operation and safety of the fuel hose evacuation kit.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Component and material inspection (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 Component and material inspection. Components and materials used in the manufacturing, testing, and packaging of the fuel hose evacuation kit shall be inspected in accordance with the applicable referenced documents. Nonconformance to the applicable referenced documents shall be cause for rejection.

4.4 First article inspection.

4.4.1 Examination. The first article shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. The first article shall be tested as specified in 4.6.2.1 through 4.6.2.3. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspections.

4.5.1 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specifed in 4.6.1. Acceptable quality level (AQL) shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each fuel hose evacuation kit shall be tested as specified in 4.6.2.1 and 4.6.2.2. Failure of test shall be cause for rejection.

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4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2.3. AQL shall be 2.5 percent defective.

4.6 Inspection procedures.

4.6.1 Examination. The fuel hose evacuation kit shall be examined for the following defects:

Major

101. Parts, components, and assemblies not free from extraneous material.
102. Castings and forgings not free from blowholes, hard spots, porosity, and cracks.
103. Castings repaired, plugged, brazed, or burned.
104. Metal fabrication not as specified.
105. Threaded connections and fasteners with evidence of cross threading and mutilation, or with detrimental or hazardous burrs.
106. Components missing or not as specifed.

Minor

201. Thread sealing compound missing or not as specified.
202. Treatment and painting or anodizing not as specified.
203. Identification plate not as specified.
204. Instruction plate not as specified.
205. Loading diagram not as specified.

4.6.2 Tests.

4.6.2.1 Evacuation test. The eductor assembly shall be tested as follows:

a. Connect the inlet coupling of the eductor assembly to an air source capable of delivering not less than 105 cubic feet of air per minute (ft^3/min) at a pressure of 80 pounds force per square inch (lb_f/in^2) plus or minus 2 lb_f/in^2.
b. Connect the evacuation coupling to a test gage capable of measuring a vacuum of 30 inches of mercury with an accuracy of 0.25-percent of full scale range.
c. Allow the air to flow through the eductor assembly at a rate of 105 to 110 ft^3/min at a pressure of 80 lb_f/in^2. Note the vacuum on the vacuum test gage at the end of 1 and 2 minutes.

Nonconformance to 3.4 shall constitute failure of this test.

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4.6.2.2 Air pressure test. The fuel hose evacuation kit shall be tested as follows:

a. Connect the sender unit assembly to the ball receiver assembly by the quick disconnect couplings.
b. Close the other end of the ball receiver assembly, using a compatible quick disconnect cap.
c. Connect the eductor assembly to the sender unit assembly, using the compatible pneumatic hose couplings.
d. Connect the remaining pneumatic hose coupling on the eductor to any suitable air source capable of maintaining an air pressure up to and including 100 lb_f/in^2.
e. Pressurize the assembly to 25 lb_f/in^2 and submerge in water. If no leaks are visible, pressurize to 50 lb_f/in^2 and again submerge in water. If no leaks are visible, increase pressurization in 10 lb_f/in^2 increments until not less than 100 lb_f/in^2 is reached. Submerge the assembly in water at the end of each increment and examine for leaks before proceeding to the next increment.
f. When assembly has been pressurized to 100 lb_f/in^2, submerge in water and hold at that pressure for not less than 5 minutes. Examine for leaks and damage.

Nonconformance to 3.4 shall constitute failure of this test.

4.6.2.3 Rough handling. The fuel hose evacuation components shall be loaded in the chest in accordance with the loading diagram. The lid shall be held closed by a clip in the hasp. The chest shall then be lifted not less than 24 inches and dropped on one corner on a surface consisting of 3 inches of packed sand over concrete. The chest and contents shall be examined for cracks, stress damage, and permanent deformation. Nicks and mars on the outside of the chest as a result of this test shall not be considered defects. Nonconformance to 3.5 shall consitute failure of this test.

4.7 Packaging inspection.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

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107. Components not placed and secured within the chest as provided for level A, B, or commercial.
108. Boxes not as specified for level A or B.
109. Unlike components packed together for level A or B.
110. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.1.1 Level A. The components shall be placed and secured within the chest provided, and the chest shall be packed in a close fitting box conforming to PPP-B-601, overseas type, style optional. When specified (see 6.2.1), chests containing like size components shall be packed together in quantities not to exceed the weight limitation of the box specified herein. Each chest shall be blocked within the box in a manner to prevent movement or damage. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.1.2 Level B. Packing shall be as specified in 5.1.1 for level A, except boxes shall be domestic type, or individually packed boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c, style optional. Box closure and strapping shall be in accordance with the appendix to the applicable box closure.

5.1.3 Commercial. The components shall be placed and secured within the chest provided. The chest shall then be packed in accordance with MIL-STD-1188.

5.2 Marking. In addition to any special marking required by the contract, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The fuel hose evacuation kits are intended for use in removing residual liquids and vapors from 4-inch and 6-inch hose lines that are components of tactical fuel systems that receive, store, transfer, and dispense gasoline, diesel fuel, and jet fuel.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

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b. Size required (see 1.2).
c. When first article is required for inspection and approval (see 3.1, 4.4, and 6.3).
d. When instruction plate is not required (see 3.5.2).
e. Level of packing required (see 5.1.1).
f. When chests are to be packed together (see 5.1.1).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4	First Article Inspection Procedures.	DI-T-4901
4.4	First Article Inspection Report.	DI-T-4902
4.5	Production/Acceptance Inspection Procedures.	DI-T-4903
4.6	Inspection System Program Plan.	DI-R-4803
4.6	End Item Final Inspection Record.	DI-R-4809

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulations. The first article should be the first-produced item. The first article should consist of one fuel hose evacuation kit. The contracting officer should include specific instructions of all procurement instruments, regarding arrangements for examination, test, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled or reclaimed material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - MC

Project 4930-N218

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DOCUMENT IDENTIFIER AND TITLE
MIL-F-82191A(MC) Fuel Hose Evacuation Kits

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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